Understanding of Diagonalizability and No Repeated Roots in Minimal Polynomial Equivalent

Problem Statement

For any matrix A, the following statements are equivalent:

1. The matrix A is diagonalizable.
2. The minimal polynomial of A has no repeated roots, meaning each linear factor has a power of 1.

1 The Extended Eigenspace and Jordan Form View

When a matrix is non-defective, we can find a full set of n eigenvector directions. This lets us split:

$$\mathbb{R}^n = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}$$

where each $E_{\lambda_i} = \text{Ker}(A - \lambda_i I)$.

However, defective matrices collapse—their eigenspaces are too small and only form a subspace of $\mathbb{R}^n$. To get the missing dimensions, we expand E_{λ_i} into the extended eigenspace, denoted T_{λ_i}.

Instead of stopping at $\text{Ker}(A - \lambda I)$, we look at vectors killed by higher powers of the operator:

- Step 1: $\text{Ker}(A - \lambda I) \rightarrow$ the core eigenvectors
- Step 2: $\text{Ker}(A - \lambda I)^2 \rightarrow$ vectors that need 2 applications to drop to $\vec{0}$
- Step m: $\text{Ker}(A - \lambda I)^m \rightarrow$ the final boundary where the kernel stops growing

The kernel space of $(A - \lambda_i I)^k$ is called T_{λ_i}, aka the extended eigenspace. Every matrix has a Jordan Form—this basically means T_{λ_i} has the same dimension as its algebraic multiplicity and thus all T_{λ_i} decompose $\mathbb{R}^n$ just like E_{λ_i}:

$$T_{\lambda_1} \oplus \cdots \oplus T_{\lambda_k} = \mathbb{R}^n$$

Connecting the Onion Ring to Jordan Blocks and Chains

To see how the Jordan Canonical Form arises naturally, we must connect the "onion ring" structure of T_{λ_i} with individual Jordan blocks:

- **Extended Eigenspace (T_{λ_i}) = The Entire Onion Ring:** The full space T_{λ_i} encompasses all nested kernel layers for eigenvalue λ_i. Its total dimension equals the algebraic multiplicity m_i.
- **1 Core Eigenvector = 1 Jordan Block:** The number of independent eigenvectors in the innermost core $\text{Ker}(A - \lambda_i I)$ equals the geometric multiplicity g_i. Each independent eigenvector serves as the terminus for **one individual Jordan chain**, which in turn yields **one Jordan block**.
- **1 Jordan Chain = 1 Path Inward:** Each chain originates in an outer layer $\text{Ker}(A - \lambda_i I)^k$ and steps radially inward toward an eigenvector in the core.

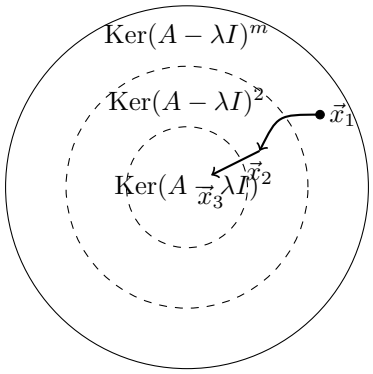
For a specific vector $\vec{x}_1 \in \text{Ker}(A-\lambda I)^3 \setminus \text{Ker}(A-\lambda I)^2$, repeatedly applying $(A-\lambda I)$ generates a chain of length 3 that terminates at an eigenvector in $\text{Ker}(A-\lambda I)$:

$$\vec{x}_1 \xrightarrow{A-\lambda I} \vec{x}_2 \xrightarrow{A-\lambda I} \vec{x}_3 \xrightarrow{A-\lambda I} \mathbf{0}$$

In our example, T_{λ_i} consists of three such chains:

$$\begin{aligned} &\vec{x}_1 \to \vec{x}_2 \to \vec{x}_3 \\ &\vec{y}_1 \to \vec{y}_2 \\ &\vec{z}_1 \end{aligned}$$

Here, T_{λ_i} has dimension 6, algebraic multiplicity 6, and geometric multiplicity 3 (corresponding to the 3 distinct chains and 3 Jordan blocks).



How Jordan Blocks Emerge from Basis Chains

Let $(\vec{m}_1, \vec{m}_2, \vec{m}_3)$ be a Jordan chain written in reverse order (where $\vec{m}_1$ is the core eigenvector in $\text{Ker}(A-\lambda I)$):

$$(A-\lambda I)\vec{m}_1 = \mathbf{0} \implies A\vec{m}_1 = \lambda\vec{m}_1$$

$$(A-\lambda I)\vec{m}_2 = \vec{m}_1 \implies A\vec{m}_2 = \vec{m}_1 + \lambda\vec{m}_2$$

$$(A-\lambda I)\vec{m}_3 = \vec{m}_2 \implies A\vec{m}_3 = \vec{m}_2 + \lambda\vec{m}_3$$

Writing this system in matrix block form gives:

$$A\begin{pmatrix}\vec{m}_1 & \vec{m}_2 & \vec{m}_3\end{pmatrix} = \begin{pmatrix}\vec{m}_1 & \vec{m}_2 & \vec{m}_3\end{pmatrix}\begin{pmatrix}\lambda & 1 & 0\\ 0 & \lambda & 1\\ 0 & 0 & \lambda\end{pmatrix}$$

This explicitly demonstrates why Jordan blocks have λ on the main diagonal and 1s on the superdiagonal: the 1s encode the step-by-step shift inward across the layers of the onion ring!

Thus, $A = MJM^{-1}$, where J is composed of block matrices on the diagonal corresponding to each chain in T_{λ_i}:

$$J = \begin{pmatrix} J_{\lambda_1} & & \\ & \ddots & \\ & & J_{\lambda_k}\end{pmatrix} \quad \text{where} \quad J_{\lambda_i} = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_{g_i}\end{pmatrix}$$

Moreover, the dim of T_{λ_i} is exactly the dim of J_{λ_i}, which is also the power of corresponding λ_i terms in characteristic functionof the linear transform. This is not yet proved here but is the directed result of the Primary Decomposition Theroem.

Thus, this Primary Decomposition Theorem unifies eigenspaces and Jordan forms: Jordan Form is simply the matrix representation of A under the extended eigenspace chain basis. If A is diagonalizable, T_{λ_i} never expands past the first ring, so $T_{\lambda_i} = E_{\lambda_i}$ for all i.

Specifically for the J matrix of the linear transform in Jordan form, we can think of J consists first a layer of J_{λ_i} block (sub matrix) on its diagonal. Each of this J_{λ_i} denotes the extended eigen space T_{λ_i}. For each J_{lambda_i}, it consists another layer of J_i on its diagonal. Each J_i denotes one chain in the onion rings of T_{λ_i}. The total number of chains is just the count of geometrical multiplicity of the eigen space (or extended eigen space) of this λ_i. This is easy to understand as each chains ends up in the most internal circle, which is just the eigen space of this λ_i, including length one chains just reside in the eigen space. Here, the internal structure of J_i is not just λ_i in the diagonal. It has also one more line of 1s in the diagonal shifted upper right one move to compensate for the chain structure as illustrated above. All in all, from the structure wise, we can see that J is made with as blocks as diagonalizable as possible. J contains the inner most blocks as J_i block. This diagonal structure has the beauty that why they multiple each other, the resulting matrix can be viewed as diagonal blocks multiplication. This would drastically make later Cayley Theorem easily to understand.

2 The Characteristic Polynomial View

To get the eigenvalues of a matrix, we search for values that satisfy $Ax = \lambda x$. This requirement is equivalent to finding $(A-\lambda I)x = \mathbf{0}$ for non-zero x, which corresponds to $\det(A-\lambda I) = 0$. This yields the characteristic polynomial:

$$P_A(\lambda) = \det(A - \lambda I) = 0$$

By the Fundamental Theorem of Algebra, $P_A(\lambda)$ factors into linear terms:

$$P_A(\lambda) = (\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2} \dots (\lambda - \lambda_k)^{m_k}$$

where m_i is the algebraic multiplicity of λ_i, matching $\dim(T_{\lambda_i})$.

3 Why Jordan Form Wipes Out the Characteristic Polynomial

3.1 The Matrix Multiplication View: Step-by-Step Block Elimination

Cayley Theorem status linear transform's matrix wipes out its characteristic polynomial. To see how a matrix annihilates its characteristic polynomial ($P_A(A) = \mathbf{0}$), we look at matrix multiplication on J. Consider a 7×7 matrix A with characteristic polynomial $P_A(x) = (x - \lambda_1)^5(x - \lambda_2)^2$.

The 5-dim space for λ_1 contains two blocks: one 3×3 block and one 2×2 block. The 2-dim space for λ_2 contains one 2×2 block. The matrix J is arranged as:

$$J = \left(\begin{array}{ccc|cc|cc} \lambda_1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & \lambda_1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & \lambda_2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_2 \end{array}\right)$$

When evaluating factors step by step, each power shifts superdiagonal 1s upward, zeroing out blocks based on their size:

1. **Eliminating the 3×3 block of λ_1 via $(J - \lambda_1 I)^3$:** Subtracting $\lambda_1 I$ zeroes out the main diagonal of the λ_1 blocks. Raising $(J - \lambda_1 I)$ to power 3 shifts the superdiagonal 1s completely off the 3×3 block, wiping it out to $\mathbf{0}$.

2. **Eliminating the 2×2 block of λ_1 via $(J - \lambda_1 I)^2$:** Raising $(J - \lambda_1 I)$ to power 2 is already enough to shift the 1s off the smaller 2×2 block of λ_1, zeroing it out completely.

3. **Matching extended eigen space dim with algebraic multiplicity** Here is the important thing. In the characteristic polynomial, the root λ_1 has exactly 5 (algebraic multiplicity) multiplicity as the above sections illustrated. This is also the dim of the extended eigen space. Thus, the number of $(J - \lambda_1 I)$ of 3 and 2 aligns with algebraic multiplicity even the geometrically multiplicity is 2. This alignment basically says the first three in item 1 eliminates the first chains in this extended eigen space. The second two in item 2 eliminates the second chains in this extended eigen space. Later this 3 by 3 0 block will make corresponding block in the end result matrix 0 block, for the reason that diagonal block matrix multiplication can be viewed as corresponding block on the diagonal multiplication. Same thing can be said for the 2 by 2 block.

4. **Eliminating the 2×2 block of λ_2 via $(J - \lambda_2 I)^2$:** Similarly, applying $(J - \lambda_2 I)^2$ zeroes out the main diagonal of the λ_2 block and shifts its superdiagonal 1s off at power 2, wiping out the entire 2×2 block of λ_2.

The Realization: Only the Longest Chain Matters

Notice what happens in the factor $(J - \lambda_1 I)^5$:

- Power 1 leaves superdiagonal 1s.
- Power 2 eliminates the smaller 2×2 block of λ_1.
- Power 3 eliminates the larger 3×3 block of λ_1.

Once the power reaches **3** (the length of the longest chain / largest block for λ_1), **all** blocks belonging to λ_1 are completely zeroed out! The extra powers up to 5 in $(J - \lambda_1 I)^5$ are entirely redundant.

This key realization reveals that to achieve annihilation, we do not need the full algebraic multiplicity m_i; we only need a power equal to the size of the *longest Jordan chain* (largest block size) for each eigenvalue!

4 Refining Cayley-Hamilton: The Emergence of the Minimal Polynomial

- **Geometric Multiplicity (g_i):** $\dim(\text{Ker}(A - \lambda_i I))$, representing the total number of independent Jordan blocks for λ_i.
- **Algebraic Multiplicity (m_i):** $\dim(T_{\lambda_i})$, the sum of sizes of all Jordan blocks for λ_i.
- Geometric and algebraic multiplicities obey $g_i \leq m_i$.

The **minimal polynomial $M_A(x)$** is defined as the unique monic polynomial of lowest degree that annihilates A, meaning $M_A(A) = M_A(J) = \mathbf{0}$. Instead of taking the total algebraic multiplicities m_i, it looks strictly at the maximum chain lengths.

For our 7×7 example, because power 3 eliminates all λ_1 blocks and power 2 eliminates all λ_2 blocks, $M_A(x)$ drops the redundant powers to achieve annihilation with a degree-5 polynomial instead of degree-7:

$$M_A(x) = (x - \lambda_1)^3 (x - \lambda_2)^2$$

5 Minimal Erase: The Natural Sequence to Diagonalizability

The Elegant Equivalence

The complete connection flows naturally:

- **Direction 1 ($\implies$):** If A is **diagonalizable**, $g_i = m_i$ for all i. Every Jordan block must be a 1×1 scalar block. The maximum chain length for every eigenvalue is 1, so the minimal polynomial has **no repeated roots**:

$$M_A(x) = (x - \lambda_1)^1 (x - \lambda_2)^1 \dots (x - \lambda_k)^1$$

- **Direction 2 ($\Longleftarrow$):** If $M_A(x)$ has **all powers equal to 1**, the maximum chain length is 1. Thus, no Jordan block can be 2×2 or larger. Every block is a 1×1 matrix, forcing $g_i = m_i$ for all eigenvalues and proving A **is diagonalizable**. $\square$